MIDWEST GEOLOGICAL
SEQUESTRATION CONSORTIUM

What is Seismic Monitoring?

Seismic Monitoring at the Illinois Basin - Decatur Project

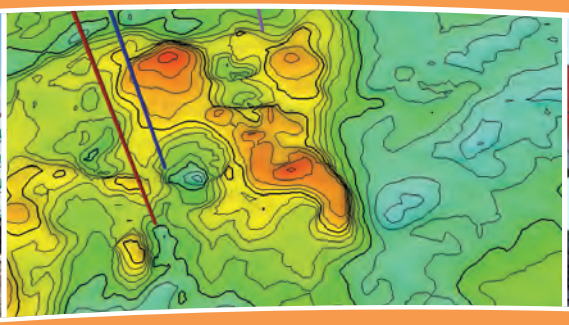
Seismic activity, or seismicity, is the energy released during the movement of rocks along a fault deep beneath the earth's surface. Magnitude represents the amount of energy released and is typically defined by the Richter scale. Earthquakes that can be felt at the surface occur when a small area of one side of a fault slides rapidly past the other side, which generates energy that transmits through the earth away from the location of the slippage. This area of slippage is the earthquake source. Earthquake motions that travel through the subsurface bedrock are many times smaller than what they are on the ground surface. When these waves travel up through the loose near-surface sediments that rest on bedrock, the waves are amplified and shaking is increased, producing the largest shaking at the ground surface. Felt seismic events, or earthquakes, are usually magnitude 2 and greater.

Comparison of Minor Global Seismic Events Organized by Richter Scale Magnitude

Magnitude	Description	Earthquake Effects	Frequency of Occurrence
<2.0	Micro	Microseismic, not felt	Continual
2.0–2.9	Very Minor	Generally not felt, but recorded	1,300,000 (est.) per year
3.0–3.9	Minor	Often felt, but rarely causes damage	130,000 (est.) per year
4.0–4.9	Light	Noticeable shaking of indoor items, rattling noises. None to slight damage	13,000 (est.) per year

Based on information from: *U.S. Geological Survey Earthquake Facts and Statistics and Magnitude/ Intensity Comparison* (http://earthquake.usgs.gov).







The earth also produces very small, unfelt, "microseismic" events that can only be recorded with highly sensitive monitoring equipment. Microseismic events are naturally occurring and may also take place in conjunction with energy-related activities, such as oil or geothermal energy production, and subsurface storage of carbon dioxide (CO_2). Microseismic events can also occur in response to minor settling a mile or more underground resulting from removing oil from the porous reservoir rocks or the small movements caused by pressure increases within the pore spaces due to injected water or CO_2.

The Midwestern portion of the United States, like all areas, experiences these small microseismic events. For example, researchers recorded 534 microseismic events in a 211-day

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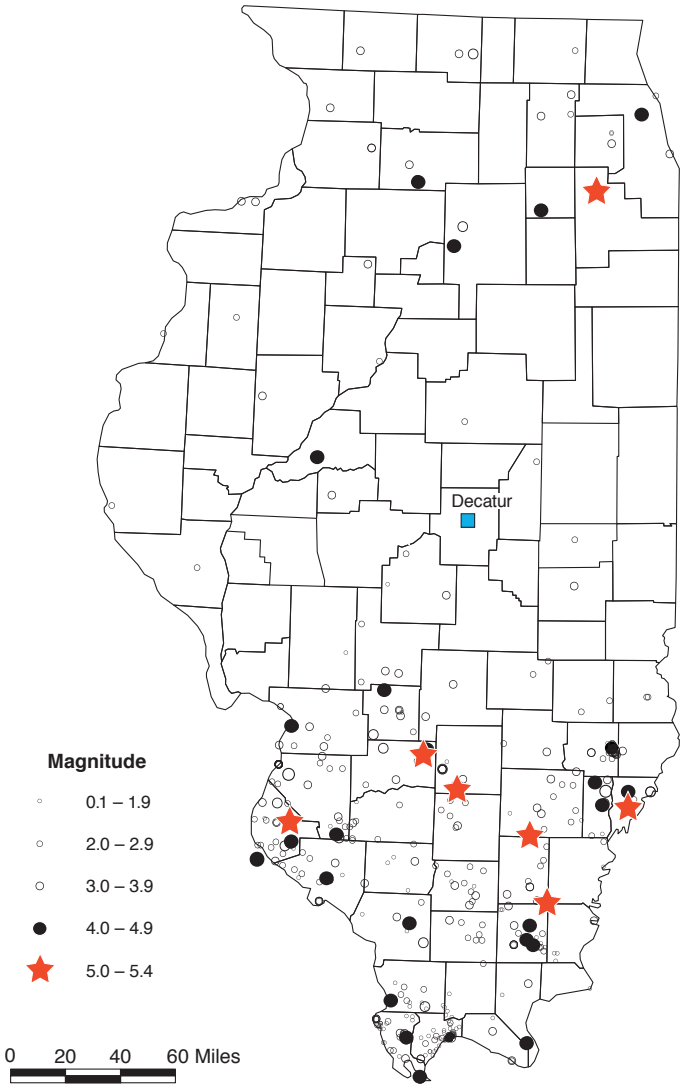
monitoring period near New Harmony, Indiana using ten closely spaced seismographs[1]. These events were likely caused by water injection wells for the purpose of enhanced recovery of oil but were too small to be felt and ranged in magnitude from 0.6 to 1.8. Vibrations similar to microseismic events can be caused by other industrial activities such as trains, large trucks, and factory operations.

The Illinois Basin – Decatur Project (IBDP) is situated in a very seismically stable area, but that does not mean no seismic activity will occur in the project area. Microseismic events can take place at any CO_2 injection site because of minor adjustments deep underground. These minor unfelt events measure far less than 2 on the Richter scale.

At the IBDP, our primary objective is the safe storage of CO_2. Therefore, along with many other types of monitoring, we are carrying out continuous microseismic and seismic monitoring at multiple locations: at the injection well itself, in a special monitoring well drilled to 3,500 feet beneath the surface, and at surface seismic stations around the project location. Subsurface monitoring began prior to injection, extends throughout the period of injection, and will continue for three years after injection stops in order to assure that we thoroughly understand any microseismic events that might be related to the project. Our monitoring allows for detection of microseismic events more than 1 billion times smaller than what would be felt at the ground surface. The Decatur site was carefully selected after extensive imaging of the subsurface showed no detectable faults in the area. The continuous monitoring undertaken throughout the project looks for any trends toward felt seismicity levels for which we have plans in place to decrease the injection rate or adjust injection pressure in response.

Earthquakes In Illinois

1795 – 2013



Historically the central part of Illinois has seen fewer recorded earthquakes than either northern or southern Illinois.

[1] Eagar, K. C., Pavlis, G. L., & Hamburger, M. W. (2006). Evidence of possible induced seismicity in the Wabash Valley seismic zone from improved microearthquake locations. *Bulletin of the Seismological Society of America,96*(5), 1718-1728.